

Homework (Carolina Reaper)

- Q1.** A music festival has two stages. The x -axis represents the frequency of the music, and the y -axis represents the volume. If the first stage operates within the region defined by $x \geq 20$ and $y \leq 50$, which of the following points is within the feasible region?
- (A) (15, 40)
(B) (25, 40)
(C) (20, 55)
(D) (25, 55)
(E) (15, 60)
- Q2.** An artist is creating a canvas with a size defined by the system of inequalities $x + y \leq 12$ and $x - y \geq -4$. What is the area of the feasible region?
- (A) 8 square units
(B) 16 square units
(C) 24 square units
(D) 32 square units
(E) 40 square units
- Q3.** A rhythm pattern is defined by the inequalities $x \geq 10$ and $y \leq 20 - x$. Which of the following points satisfies both inequalities?
- (A) (5, 15)
(B) (10, 10)
(C) (15, 5)
(D) (20, 0)
(E) (25, -5)
- Q4.** A sound wave has a frequency defined by the inequality $y \geq 2x - 5$. If the volume is defined by the inequality $y \leq -x + 10$, which of the following points is within the overlapping region?
- (A) (0, 5)
(B) (2, 4)
(C) (4, 3)
(D) (6, 2)
(E) (8, 1)
- Q5.** A painter is mixing colors on a palette. The amount of red paint is defined by $x \geq 5$, and the amount of blue paint is defined by $y \leq 15 - x$. Which of the following points is within the feasible region?
- (A) (0, 10)
(B) (5, 10)
(C) (10, 5)
(D) (15, 0)
(E) (20, -5)
- Q6.** A music producer is working on a new track. The tempo is defined by $x \geq 100$, and the duration is defined by $y \leq 200 - x$. Which of the following points satisfies both inequalities?
- (A) (50, 150)
(B) (100, 100)
(C) (150, 50)
(D) (200, 0)
(E) (250, -50)
- Q7.** A musician is playing a melody on a 4-string instrument. The notes are defined by the system of inequalities $x + y \leq 8$ and $x - y \geq -2$. What is the area of the feasible region?
- (A) 6 square units
(B) 8 square units
(C) 10 square units
(D) 12 square units
(E) 16 square units
- Q8.** A light designer is creating a lighting scheme for a concert. The intensity of the lights is defined by $y \geq x - 3$, and the color temperature is defined by $y \leq -x + 12$. Which of the following points is within the overlapping region?
- (A) (0, 9)
(B) (3, 6)
(C) (6, 3)
(D) (9, 0)
(E) (12, -3)

Open Ended Questions (FRQ)

- Q9.** Graph the system of inequalities $x + y \leq 10$ and $x - y \geq -2$ on the same coordinate plane. Identify the overlapping solution region and interpret the feasible region in the context of a music festival.
- Q10.** A sound engineer is working on a live performance. The sound levels are defined by the system of inequalities $y \geq 2x - 5$ and $y \leq -x + 15$. Graph the system of inequalities and identify the overlapping solution region. Interpret the feasible region in the context of the live performance.

Chef's Tips

- Tip 1: Ensure students understand the concept of feasible regions and how to graph multiple inequalities on the same coordinate plane.
- Tip 2: Emphasize the importance of identifying overlapping solution regions and interpreting the feasible region in the context of the problem.
- Tip 3: Encourage students to use real-world examples, such as music festivals or live performances, to illustrate their understanding of the concepts.